SOMAIYA VIDYAVIHAR UNIVERSITY
K. J. Somaiya School of Engineering, Mumbai -77
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Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 1

Experiment: 1

Title: Simulation of Multi Server System: Able – Baker Carhop Problem.

Problem Statement: Consider a drive in restaurant where carhops take order and bring food to the cars. Cars arrive in manner as shown:

Time between Arrival(minutes)	1	2	3	4
Probability	0.25	0.4	0.2	0.15

There are 2 carhops Able & Baker. Able is better to do the job and works a bit faster than Baker. Their service distribution is as follows:

Service Time(minutes)	2	3	4	5
Probability	0.3	0.28	0.25	0.17

Service Distribution time of Baker:

Service time(minutes)	3	4	5	6
Probability	0.35	0.25	0.2	0.2

Able gets the customer if both carhops are idle. The problem is to find how well the current arrangement is working.

Expected Outcome of Experiment:

Index	Outcome
CO1	Understand the concepts of discrete event simulation and its importance in business, science, engineering, industry and other services.

Books/ Journals/ Websites referred:

1. Jerry Banks, John Carson, Barry Nelson, and David M. Nicol, "Discrete Event System Simulation,; Fifth Edition, Prentice-Hall.
2. Averill M Law, "System Modeling & Analysis"; 4th Edition TMH.
3. Banks C M, Sokolowski J A, "Principles of Modeling and Simulation", Wiley

Pre Lab/ Prior Concepts:

Theory:

Conceptual Model:

I. Discrete event model of system used for multichannel queuing. E.g. of Able & baker problem.
II. This problem is simulated using an event scheduling simulation.
III. A simulation table is used to record the excessive system snapshot as time proceeds.
IV. The simulation requires mainly an activity table representing a service time distribution of able & baker & inters arrival of customers.
V. Activity duration is specified by a modeller.

Characteristics of System:

I) Calling Population: Infinite in nature.
II) System capacity: Infinite.
III) Nature of Arrival: Random arrival nature.
IV) Service Mechanism: At a time maximum two customers can be served one by Able & other by Baker. If able & baker both are busy, the customer has to wait. If both servers are free, priority goes to Able.
V) Queuing Discipline: Customers are chosen in FIFO manners.

System State:

System state for Able or Baker indicating Able being Idle or Busy at given instant.

Entities:
Neither the customers nor the server needs to be explicitly represented except in

terms of state variable unless customer averages are desired.

Events:

I. Arrival Event
II. Service Completion by Able
III. Service completion by Baker.

Delay:

A customer waits in queue until Able or Baker becomes free.

Use of Random Numbers:

I. To generate random nos. in simulation packages, RAND ( ) of function is used.
II. In Able & Baker problem random nos. are used for arranging inter arrival timer & service required for customers.

Real time example:

I. Public Telephone Booth with Two Telephones
II. Customers are chosen in FIFO manner.

Result: (Performance Measures):

Average Waiting Time = (Total time customers wait in queue) / (Total no. of Customers) = **0.533**

Prob. of Customers waiting = (No. of Customers who waits) / (Total no. of Customers) = **0.04**

Prob. of Idle Server = (Total Idle Time of Server) / (Total runtime of simulation) = **16/66**

Average Service Time = (Total Service Time) / (Total no. of Customers) = **3.8667**

Average Time between Arrival=(Total Time between arrivals)/(No. of arrivals)= **2.357142857**

Average Waiting Time of Those Who Wait = (Total Time Customer waits in system) / (Total no. of Customers) = **2.0000**

Average Time Customers Spends in System = (Total Time Customer spends in system) / (Total no. of Customers) = **4.40**

Average Service Time of Able	3.7778
Average Service Time of Baker	4.0000

Conclusion:

Thus, from this experiment, we observe that the Able–Baker system operates efficiently with low customer wait times and balanced server utilization.

Post Lab Questions:

Plot the frequency of caller delay & average caller delay for 30 trials.
ANS.: Attached